

Brutal Paper B28 - Winds, Storms & Cyclones | Respiration in Organisms | Perimeter & Area | Lines & Angles

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A rectangle is 18 cm long and 12 cm wide. A path 2 cm wide runs INSIDE, all along the boundary. What is the area of the path? [\[Perimeter & Area\]](#)
 - (A) 104 cm²
 - (B) 96 cm²
 - (C) 60 cm²
 - (D) 120 cm²
 - (E) 108 cm²
 - (F) 100 cm²
 - (G) 112 cm²

2. Two angles are complementary and one of them is 4 times the other. Find the larger angle. [\[Lines & Angles\]](#)
 - (A) 108 deg
 - (B) 144 deg
 - (C) 75 deg
 - (D) 72 deg
 - (E) 54 deg
 - (F) 36 deg
 - (G) 18 deg

3. A cyclone's wind blows at 180 km/h. Express this speed in metres per second (multiply by 5/18). [\[Winds, Storms & Cyclones\]](#)
 - (A) 25 m/s
 - (B) 18 m/s
 - (C) 50 m/s
 - (D) 60 m/s
 - (E) 90 m/s
 - (F) 30 m/s
 - (G) 648 m/s

4. A person at rest breathes 18 times each minute. How many breaths is that in one full hour? [\[Respiration in Organisms\]](#)
 - (A) 60
 - (B) 720
 - (C) 1800
 - (D) 900
 - (E) 180
 - (F) 1080
 - (G) 108

5. A circle is drawn touching all four sides of a square of side 14 cm. Using $\pi = \frac{22}{7}$, find the area left inside the square but outside the circle. [\[Perimeter & Area\]](#)
- (A) 100 cm²
 - (B) 42 cm²
 - (C) 84 cm²
 - (D) 196 cm²
 - (E) 49 cm²
 - (F) 38.5 cm²
6. Two supplementary angles are in the ratio 2 : 3. Find the smaller angle. [\[Lines & Angles\]](#)
- (A) 72 deg
 - (B) 60 deg
 - (C) 36 deg
 - (D) 108 deg
 - (E) 45 deg
 - (F) 120 deg
 - (G) 90 deg
7. Before a storm a barometer reading falls from 75 cm of mercury to 72 cm. What is the percentage drop in pressure? [\[Winds, Storms & Cyclones\]](#)
- (A) 2.4%
 - (B) 6%
 - (C) 4%
 - (D) 3%
 - (E) 4.16%
 - (F) 12%
8. During exercise a person's breathing rate rises from 15 to 25 breaths per minute. What is the percentage increase? [\[Respiration in Organisms\]](#)
- (A) 66.67%
 - (B) 40%
 - (C) 25%
 - (D) 10%
 - (E) 166.7%
 - (F) 50%
 - (G) 60%
9. A rectangular field is 80 m by 60 m. Fencing costs Rs 25 per metre. What is the total cost of fencing it once around? [\[Perimeter & Area\]](#)
- (A) Rs 14000
 - (B) Rs 7000
 - (C) Rs 6000
 - (D) Rs 3500
 - (E) Rs 12000
 - (F) Rs 8000
 - (G) Rs 2800

10. An angle's supplement is 30 degrees more than twice its complement. Find the angle. [\[Lines & Angles\]](#)

- (A) 45 deg
- (B) 90 deg
- (C) 40 deg
- (D) 60 deg
- (E) 20 deg
- (F) 150 deg
- (G) 30 deg

11. At 2 pm a seaside village finds its land far warmer than the nearby water. Which way does the breeze flow? [\[Winds, Storms & Cyclones\]](#)

- (A) There is no breeze
- (B) From north to south
- (C) From east to west
- (D) From sea to land
- (E) From land to sea
- (F) From land to land
- (G) Straight upward only

12. During very heavy exercise, when oxygen runs short, our muscle cells respire anaerobically and produce: [\[Respiration in Organisms\]](#)

- (A) Only water
- (B) Starch
- (C) Alcohol and carbon dioxide
- (D) Lactic acid
- (E) Glucose and oxygen
- (F) Carbon monoxide
- (G) Pure oxygen

13. A square and a rectangle have equal perimeters. The square has side 16 cm and the rectangle is 20 cm long. Find the area of the rectangle. [\[Perimeter & Area\]](#)

- (A) 160 cm²
- (B) 320 cm²
- (C) 256 cm²
- (D) 240 cm²
- (E) 400 cm²
- (F) 200 cm²

14. Two parallel lines are cut by a transversal. One of a pair of co-interior (same-side) angles is 65 degrees. Find the other. [\[Lines & Angles\]](#)

- (A) 105 deg
- (B) 115 deg
- (C) 125 deg
- (D) 130 deg
- (E) 90 deg
- (F) 25 deg
- (G) 65 deg

- 15.** A cyclone's eye is 30 km across and the whole storm is moving at 15 km/h. How long does the calm eye take to pass completely over a town? [\[Winds, Storms & Cyclones\]](#)
- (A) 45 h
 - (B) 15 h
 - (C) 2 h
 - (D) 3 h
 - (E) 1 h
 - (F) 4 h
 - (G) 0.5 h
- 16.** When yeast respire anaerobically (without oxygen), the products are: [\[Respiration in Organisms\]](#)
- (A) Alcohol and carbon dioxide
 - (B) Lactic acid only
 - (C) Oxygen and energy
 - (D) Water and oxygen
 - (E) Only carbon dioxide
 - (F) Vinegar and water
 - (G) Glucose and starch
- 17.** A floor 5 m by 4 m is to be tiled with tiles measuring 50 cm by 25 cm. How many whole tiles are needed? [\[Perimeter & Area\]](#)
- (A) 200
 - (B) 160
 - (C) 40
 - (D) 16
 - (E) 100
 - (F) 320
- 18.** The two angles of a linear pair are $(2x + 10)$ degrees and $(3x - 30)$ degrees. Find x . [\[Lines & Angles\]](#)
- (A) 44
 - (B) 50
 - (C) 38
 - (D) 40
 - (E) 36
 - (F) 20
 - (G) 30
- 19.** What happens to a parcel of air when it is warmed? [\[Winds, Storms & Cyclones\]](#)
- (A) It contracts and sinks
 - (B) It turns into water at once
 - (C) It stops moving completely
 - (D) It expands, becomes lighter and rises
 - (E) It gets denser and falls
 - (F) It freezes instantly

20. During inhalation (breathing in), what does the diaphragm do? [\[Respiration in Organisms\]](#)

- (A) It blocks the windpipe
- (B) It pushes air out forcefully
- (C) It squeezes the lungs smaller
- (D) It moves down and flattens, increasing the chest cavity
- (E) It dissolves into the lungs
- (F) It moves up and domes
- (G) It stays completely still

21. The circumference of a circular plate is 44 cm. Using $\pi = \frac{22}{7}$, find its area. [\[Perimeter & Area\]](#)

- (A) 22 cm²
- (B) 308 cm²
- (C) 49 cm²
- (D) 154 cm²
- (E) 44 cm²
- (F) 77 cm²

22. Three angles meet at a point and are 90 degrees, 120 degrees, and x. Find x. [\[Lines & Angles\]](#)

- (A) 90 deg
- (B) 180 deg
- (C) 210 deg
- (D) 130 deg
- (E) 150 deg
- (F) 160 deg

23. A steady wind covers 720 km in 8 hours. What is its speed? [\[Winds, Storms & Cyclones\]](#)

- (A) 25 km/h
- (B) 80 km/h
- (C) 9 km/h
- (D) 120 km/h
- (E) 72 km/h
- (F) 5760 km/h
- (G) 90 km/h

24. A person inhales 500 mL of air per breath and takes 16 breaths each minute. How much air passes through the lungs in one minute? [\[Respiration in Organisms\]](#)

- (A) 8000 mL
- (B) 16000 mL
- (C) 800 mL
- (D) 1600 mL
- (E) 80000 mL
- (F) 500 mL

25. A wire 88 cm long is bent into a perfect circle. Using $\pi = \frac{22}{7}$, find the area enclosed. [\[Perimeter & Area\]](#)

- (A) 1232 cm²
- (B) 44 cm²
- (C) 154 cm²
- (D) 308 cm²
- (E) 88 cm²
- (F) 616 cm²
- (G) 196 cm²

26. Two vertically opposite angles are $(3x + 5)$ degrees and $(2x + 25)$ degrees. Find the measure of each angle. [\[Lines & Angles\]](#)

- (A) 45 deg
- (B) 70 deg
- (C) 20 deg
- (D) 85 deg
- (E) 25 deg
- (F) 65 deg

27. A wind that blows FROM the west toward the east is called a: [\[Winds, Storms & Cyclones\]](#)

- (A) Easterly
- (B) Northerly
- (C) Downwind
- (D) Westerly
- (E) Upwind
- (F) Crosswind

28. Compared with the air we breathe in, the air we breathe OUT contains a larger amount of: [\[Respiration in Organisms\]](#)

- (A) Helium
- (B) Oxygen
- (C) Dust only
- (D) Water vapour only
- (E) Nitrogen
- (F) Hydrogen
- (G) Carbon dioxide

29. A rectangle has a perimeter of 60 cm and its length is twice its breadth. Find its area. [\[Perimeter & Area\]](#)

- (A) 180 cm²
- (B) 300 cm²
- (C) 225 cm²
- (D) 150 cm²
- (E) 200 cm²
- (F) 400 cm²
- (G) 100 cm²

30. An angle is equal to twice its own supplement. Find the angle. [\[Lines & Angles\]](#)

- (A) 60 deg
- (B) 45 deg
- (C) 40 deg
- (D) 90 deg
- (E) 120 deg
- (F) 135 deg
- (G) 150 deg

31. During an experiment, air is pumped OUT of a sealed tin can and the can crumples inward. Why? [\[Winds, Storms & Cyclones\]](#)

- (A) Heat melted the can
- (B) The metal dissolved
- (C) Gravity pulled it down
- (D) The can gained extra air
- (E) Inside air froze and shrank
- (F) Outside air pressure is now greater than inside
- (G) The can lost weight suddenly

32. Through which structures does a cockroach take in air for respiration? [\[Respiration in Organisms\]](#)

- (A) A single nostril
- (B) Feathery lungs
- (C) Spiracles leading to air tubes (tracheae)
- (D) Moist skin
- (E) Lungs
- (F) Gills
- (G) Its mouth only

33. A path 1 m wide is laid all around the OUTSIDE of a rectangular garden 10 m by 6 m. Find the area of the path. [\[Perimeter & Area\]](#)

- (A) 16 m²
- (B) 40 m²
- (C) 32 m²
- (D) 36 m²
- (E) 28 m²
- (F) 60 m²
- (G) 96 m²

34. Two parallel lines are cut by a transversal. An angle of $(3x)$ degrees and its corresponding angle of $(x + 50)$ degrees are formed. Find the size of the angle. [\[Lines & Angles\]](#)

- (A) 60 deg
- (B) 50 deg
- (C) 25 deg
- (D) 150 deg
- (E) 75 deg
- (F) 105 deg
- (G) 15 deg

- 35.** A cyclone warning is issued 6 hours before landfall, when the storm is 240 km from the coast. At what speed is the storm approaching? [\[Winds, Storms & Cyclones\]](#)
- (A) 1440 km/h
 - (B) 30 km/h
 - (C) 20 km/h
 - (D) 24 km/h
 - (E) 40 km/h
 - (F) 48 km/h
- 36.** How do fish obtain oxygen for respiration? [\[Respiration in Organisms\]](#)
- (A) They use gills to take in oxygen dissolved in water
 - (B) They use spiracles like insects
 - (C) They swallow air bubbles only
 - (D) They breathe through fins
 - (E) They use lungs to breathe air
 - (F) They absorb oxygen through scales
 - (G) They make oxygen inside the body
- 37.** The area of a triangle is 84 cm² and its base is 14 cm. Find its height. [\[Perimeter & Area\]](#)
- (A) 6 cm
 - (B) 24 cm
 - (C) 8 cm
 - (D) 7 cm
 - (E) 168 cm
 - (F) 10 cm
 - (G) 12 cm
- 38.** Two angles form a linear pair and one is 24 degrees less than the other. Find the larger angle. [\[Lines & Angles\]](#)
- (A) 66 deg
 - (B) 90 deg
 - (C) 108 deg
 - (D) 102 deg
 - (E) 84 deg
 - (F) 96 deg
 - (G) 78 deg
- 39.** Which combination of conditions describes a cyclone? [\[Winds, Storms & Cyclones\]](#)
- (A) Low temperature with light snow
 - (B) Steady pressure with mild breeze
 - (C) High pressure with gentle winds
 - (D) Low pressure with very high-speed winds and heavy rainfall
 - (E) High pressure with calm, dry air
 - (F) High pressure with heavy rain only
 - (G) No wind with clear skies

40. How does an earthworm carry out gas exchange? [\[Respiration in Organisms\]](#)

- (A) Through its moist skin
- (B) Through gills
- (C) Through spiracles
- (D) Through a beak
- (E) Through dry scales
- (F) Through its tail only

41. A square has an area of 169 cm². What is its perimeter? [\[Perimeter & Area\]](#)

- (A) 104 cm
- (B) 52 cm
- (C) 676 cm
- (D) 338 cm
- (E) 42 cm
- (F) 26 cm
- (G) 13 cm

42. Two complementary angles differ by 14 degrees. Find the larger angle. [\[Lines & Angles\]](#)

- (A) 62 deg
- (B) 45 deg
- (C) 76 deg
- (D) 28 deg
- (E) 66 deg
- (F) 52 deg
- (G) 38 deg

43. A weather station needs a reading of how fast the wind is blowing. Which instrument gives this? [\[Winds, Storms & Cyclones\]](#)

- (A) Barometer
- (B) Weather cock
- (C) Hygrometer
- (D) Rain gauge
- (E) Thermometer
- (F) Anemometer
- (G) Lactometer

44. When exhaled air is passed through lime water, the lime water turns milky. This proves the exhaled air contains: [\[Respiration in Organisms\]](#)

- (A) Helium
- (B) Water only
- (C) Carbon dioxide
- (D) Hydrogen
- (E) Dust
- (F) Nitrogen
- (G) Oxygen

45. From a rectangular sheet 30 cm by 20 cm, a square of side 5 cm is cut from EACH of the four corners. What area of sheet remains? [\[Perimeter & Area\]](#)
- (A) 500 cm²
 - (B) 450 cm²
 - (C) 550 cm²
 - (D) 400 cm²
 - (E) 575 cm²
 - (F) 100 cm²
46. What is the reflex angle corresponding to an angle of 110 degrees? [\[Lines & Angles\]](#)
- (A) 110 deg
 - (B) 70 deg
 - (C) 250 deg
 - (D) 160 deg
 - (E) 290 deg
 - (F) 180 deg
 - (G) 360 deg
47. As you climb higher up a tall mountain, what happens to the air pressure? [\[Winds, Storms & Cyclones\]](#)
- (A) It decreases
 - (B) It becomes zero at once
 - (C) It first rises then vanishes
 - (D) It triples with height
 - (E) It increases
 - (F) It stays exactly the same
48. When do green plants carry out respiration? [\[Respiration in Organisms\]](#)
- (A) Only at night
 - (B) All the time, both day and night
 - (C) Only once a week
 - (D) Only during the day
 - (E) Only when flowering
 - (F) Never, plants do not respire
49. Two squares have sides in the ratio 2 : 3. What is the ratio of their areas? [\[Perimeter & Area\]](#)
- (A) 8 : 27
 - (B) 16 : 81
 - (C) 4 : 9
 - (D) 2 : 3
 - (E) 9 : 4
 - (F) 6 : 4
 - (G) 4 : 6

50. What is the angle between the hour and minute hands of a clock at exactly 4 o'clock? [\[Lines & Angles\]](#)

- (A) 140 deg
- (B) 60 deg
- (C) 100 deg
- (D) 90 deg
- (E) 150 deg
- (F) 120 deg

51. A storm dumps 120 mm of rain in 5 hours. What is the average rainfall rate? [\[Winds, Storms & Cyclones\]](#)

- (A) 600 mm/h
- (B) 12 mm/h
- (C) 5 mm/h
- (D) 24 mm/h
- (E) 20 mm/h
- (F) 48 mm/h
- (G) 30 mm/h

52. Which shows the correct path of air during inhalation? [\[Respiration in Organisms\]](#)

- (A) Nostrils -> bronchi -> windpipe -> lungs
- (B) Nostrils -> lungs -> windpipe -> bronchi
- (C) Nostrils -> nasal cavity -> windpipe -> bronchi -> lungs
- (D) Bronchi -> nostrils -> lungs -> windpipe
- (E) Windpipe -> nostrils -> lungs -> bronchi
- (F) Lungs -> bronchi -> windpipe -> nostrils

53. A circle has radius 21 cm. Using $\pi = \frac{22}{7}$, find the area of a QUARTER of the circle. [\[Perimeter & Area\]](#)

- (A) 173.25 cm²
- (B) 1386 cm²
- (C) 33 cm²
- (D) 346.5 cm²
- (E) 132 cm²
- (F) 693 cm²
- (G) 231 cm²

54. The bisector of an angle splits it into two equal parts of 35 degrees each. Find the supplement of the original angle. [\[Lines & Angles\]](#)

- (A) 70 deg
- (B) 90 deg
- (C) 140 deg
- (D) 55 deg
- (E) 110 deg
- (F) 35 deg
- (G) 145 deg

55. During a storm, the roof of a hut is lifted off even though nothing pushes it from below. The best explanation is: [\[Winds, Storms & Cyclones\]](#)

- (A) Lightning struck the roof
- (B) The roof expanded in the heat
- (C) Fast wind above the roof lowers the pressure there, so higher pressure below pushes the roof up
- (D) The roof became lighter in the rain
- (E) Gravity reversed during the storm
- (F) The wind blew underneath and lifted it
- (G) Rain dissolved the roof's weight

56. A factory worker breathes 20 times per minute. How many breaths does he take during an 8-hour shift?

[\[Respiration in Organisms\]](#)

- (A) 4800
- (B) 160
- (C) 480
- (D) 19200
- (E) 9600
- (F) 960

57. A wall is 8 m long and 5 m high. Painting costs Rs 35 per square metre. Find the cost to paint it. [\[Perimeter](#)

[& Area\]](#)

- (A) Rs 280
- (B) Rs 1750
- (C) Rs 2400
- (D) Rs 700
- (E) Rs 910
- (F) Rs 1400

58. Two angles form a linear pair and one is 5 times the other. Find the smaller angle. [\[Lines & Angles\]](#)

- (A) 25 deg
- (B) 60 deg
- (C) 150 deg
- (D) 36 deg
- (E) 30 deg
- (F) 18 deg
- (G) 45 deg

59. At night near the coast, the land cools faster than the sea. In which direction does the breeze blow?

[\[Winds, Storms & Cyclones\]](#)

- (A) From sea to sea
- (B) From east to west
- (C) No breeze at night
- (D) Straight upward
- (E) From land to sea
- (F) From sea to land
- (G) From south to north

60. After a sprint, the build-up of lactic acid in muscles causes cramps. This lactic acid forms because of:

[Respiration in Organisms]

- (A) Excess water in the blood
- (B) Lack of any food
- (C) Too little carbon dioxide
- (D) Extra sleep
- (E) Too much oxygen in the muscles
- (F) Insufficient oxygen supply to the muscles

61. A square plot has a side of 25 m. A man walks all the way around it 4 times. What total distance does he walk? [Perimeter & Area]

- (A) 400 m
- (B) 200 m
- (C) 2500 m
- (D) 625 m
- (E) 800 m
- (F) 625 m²

62. Three angles lie on a straight line and measure x , $2x$, and $3x$. Find the largest of them. [Lines & Angles]

- (A) 120 deg
- (B) 75 deg
- (C) 30 deg
- (D) 90 deg
- (E) 180 deg
- (F) 60 deg

63. A wind blows at 25 m/s. Express this speed in km/h (multiply by $18/5$). [Winds, Storms & Cyclones]

- (A) 45 km/h
- (B) 6.94 km/h
- (C) 50 km/h
- (D) 125 km/h
- (E) 90 km/h
- (F) 72 km/h

64. During exhalation (breathing out), what happens to the diaphragm and chest cavity? [Respiration in Organisms]

- (A) The diaphragm moves up and the chest cavity becomes smaller
- (B) The chest cavity grows larger
- (C) Both stay completely still
- (D) The chest fills with more air
- (E) The diaphragm contracts downward
- (F) The diaphragm flattens and the chest enlarges
- (G) The diaphragm leaves the body

65. A rectangular tile is 1.2 m long and 80 cm wide. Find its area in square centimetres. [\[Perimeter & Area\]](#)

- (A) 96 cm²
- (B) 12000 cm²
- (C) 9600 cm²
- (D) 9.6 cm²
- (E) 400 cm²
- (F) 800 cm²
- (G) 960 cm²

66. Two parallel lines are cut by a transversal. The co-interior angles are $(x + 20)$ degrees and $(2x + 10)$ degrees. Find the larger angle. [\[Lines & Angles\]](#)

- (A) 100 deg
- (B) 90 deg
- (C) 60 deg
- (D) 110 deg
- (E) 70 deg
- (F) 130 deg

67. What is the main source of energy that powers a tropical cyclone? [\[Winds, Storms & Cyclones\]](#)

- (A) Heat from underground rocks
- (B) Cold ocean currents
- (C) Sunlight reflected off ice caps
- (D) Friction from the rotating Earth alone
- (E) Heat released by water vapour from warm ocean water
- (F) Cold dry air over deserts
- (G) Light from the moon

68. In aerobic respiration, glucose combines with oxygen. The products are: [\[Respiration in Organisms\]](#)

- (A) Lactic acid only
- (B) Carbon dioxide, water and energy
- (C) Hydrogen and water
- (D) Alcohol and carbon dioxide
- (E) Nitrogen and energy
- (F) Only oxygen and glucose

69. A parallelogram has a base of 18 cm and a corresponding height of 9 cm. Find its area. [\[Perimeter & Area\]](#)

- (A) 162 cm²
- (B) 144 cm²
- (C) 54 cm²
- (D) 180 cm²
- (E) 81 cm²
- (F) 324 cm²
- (G) 27 cm²

70. An angle is 20 degrees more than its own complement. Find the angle. [\[Lines & Angles\]](#)

- (A) 55 deg
- (B) 65 deg
- (C) 70 deg
- (D) 50 deg
- (E) 45 deg
- (F) 110 deg
- (G) 35 deg

71. Which instrument shows the DIRECTION of the wind? [\[Winds, Storms & Cyclones\]](#)

- (A) Rain gauge
- (B) Hygrometer
- (C) Weather cock (wind vane)
- (D) Speedometer
- (E) Anemometer
- (F) Barometer

72. In an insect, the tracheal tubes carry air directly to the: [\[Respiration in Organisms\]](#)

- (A) Skin surface only
- (B) Blood, which then carries it
- (C) Heart only
- (D) Brain only
- (E) Mouth only
- (F) Gills
- (G) Body cells and tissues

73. A path 2 m wide runs straight across the middle of a 30 m by 20 m field, parallel to the 20 m side. Find the area of this path. [\[Perimeter & Area\]](#)

- (A) 46 m²
- (B) 30 m²
- (C) 40 m²
- (D) 52 m²
- (E) 60 m²
- (F) 100 m²
- (G) 80 m²

74. Five equal angles meet at a point and together make a full turn. Find each angle. [\[Lines & Angles\]](#)

- (A) 80 deg
- (B) 60 deg
- (C) 90 deg
- (D) 45 deg
- (E) 75 deg
- (F) 36 deg
- (G) 72 deg

75. An inflated balloon keeps its rounded shape because the air inside: [\[Winds, Storms & Cyclones\]](#)

- (A) Has no pressure at all
- (B) Pushes equally outward in all directions
- (C) Only pushes downward
- (D) Sticks the walls together
- (E) Pulls the walls inward
- (F) Pushes inward on the walls
- (G) Pushes only upward

76. A young child breathes about 24 times per minute. How many breaths is that in 15 minutes? [\[Respiration in Organisms\]](#)

- (A) 24
- (B) 36
- (C) 1440
- (D) 480
- (E) 360
- (F) 160
- (G) 240

77. A right-angled triangle has the two sides forming the right angle equal to 3 cm and 4 cm (hypotenuse 5 cm). Find its area. [\[Perimeter & Area\]](#)

- (A) 5 cm²
- (B) 7.5 cm²
- (C) 60 cm²
- (D) 12 cm²
- (E) 6 cm²
- (F) 30 cm²

78. Two adjacent angles on a straight line are $(4x)$ degrees and $(5x)$ degrees. Find the bigger angle. [\[Lines & Angles\]](#)

- (A) 140 deg
- (B) 90 deg
- (C) 160 deg
- (D) 100 deg
- (E) 120 deg
- (F) 60 deg

79. What best describes the centre, or 'eye', of a cyclone? [\[Winds, Storms & Cyclones\]](#)

- (A) A calm, low-pressure region with light winds
- (B) The stormiest, windiest part
- (C) A zone of strong upward lightning
- (D) A pocket of very high pressure
- (E) A solid spinning core of ice
- (F) A region of high pressure and dry air
- (G) The place with the heaviest rain

80. Respiration is important because it releases the energy stored in: [\[Respiration in Organisms\]](#)

- (A) Water alone
- (B) Soil minerals
- (C) Food (glucose)
- (D) Air pressure
- (E) Cold temperatures
- (F) Sunlight directly
- (G) Sound waves

81. A rectangle has the same area as a square of side 12 cm. If the rectangle's breadth is 8 cm, find its length. [\[Perimeter & Area\]](#)

- (A) 12 cm
- (B) 20 cm
- (C) 9 cm
- (D) 144 cm
- (E) 16 cm
- (F) 18 cm
- (G) 24 cm

82. The complement of an angle is one-fourth of its supplement. Find the angle. [\[Lines & Angles\]](#)

- (A) 75 deg
- (B) 50 deg
- (C) 120 deg
- (D) 30 deg
- (E) 60 deg
- (F) 90 deg

83. A cyclone travels 270 km in 6 hours. What is its speed in metres per second (use $1 \text{ km/h} = \frac{5}{18} \text{ m/s}$)?

[\[Winds, Storms & Cyclones\]](#)

- (A) 45 m/s
- (B) 9 m/s
- (C) 30 m/s
- (D) 25 m/s
- (E) 12.5 m/s
- (F) 15 m/s

84. Why do we breathe faster and more deeply just after running hard? [\[Respiration in Organisms\]](#)

- (A) To reduce energy use
- (B) Muscles need more oxygen to release more energy
- (C) We need to cool the blood only
- (D) To take in more nitrogen
- (E) To remove water from the body
- (F) The lungs have shrunk
- (G) Because the heart has stopped

85. A rectangle's length is twice its breadth and its area is 72 cm². Find its perimeter. [[Perimeter & Area](#)]

- (A) 18 cm
- (B) 48 cm
- (C) 72 cm
- (D) 144 cm
- (E) 24 cm
- (F) 36 cm
- (G) 12 cm

86. A clock shows exactly half past three (3:30). Find the angle between its hour and minute hands. [[Lines & Angles](#)]

- (A) 75 deg
- (B) 180 deg
- (C) 15 deg
- (D) 60 deg
- (E) 105 deg
- (F) 90 deg
- (G) 45 deg

87. Why are cyclonic winds so destructive to buildings? [[Winds, Storms & Cyclones](#)]

- (A) They blow only at night
- (B) They are extremely cold
- (C) They are perfectly silent
- (D) They contain no air
- (E) They move very slowly
- (F) Their very high speed lets them exert a large force on structures
- (G) They carry no moisture at all

88. A patient's breathing rate falls from 30 to 18 breaths per minute after resting. What is the percentage decrease? [[Respiration in Organisms](#)]

- (A) 66.67%
- (B) 30%
- (C) 60%
- (D) 48%
- (E) 12%
- (F) 20%
- (G) 40%

89. A circular park has a radius of 14 m. A fence is put around it at Rs 10 per metre ($\pi = 22/7$). Find the cost. [[Perimeter & Area](#)]

- (A) Rs 880
- (B) Rs 6160
- (C) Rs 616
- (D) Rs 440
- (E) Rs 1540
- (F) Rs 1760
- (G) Rs 88

- 90.** Two straight lines intersect and one of the four angles formed is 40 degrees. Find the largest of the four angles. [\[Lines & Angles\]](#)
- (A) 160 deg
 - (B) 140 deg
 - (C) 320 deg
 - (D) 100 deg
 - (E) 220 deg
 - (F) 180 deg
 - (G) 40 deg
- 91.** A coastal district recorded 8 cyclones over 40 years. On average, how many cyclones is that per decade? [\[Winds, Storms & Cyclones\]](#)
- (A) 2
 - (B) 5
 - (C) 20
 - (D) 0.2
 - (E) 4
 - (F) 8
 - (G) 16
- 92.** When you hold your breath, the urge to breathe again is triggered mainly by the rising level of: [\[Respiration in Organisms\]](#)
- (A) Heat in the chest
 - (B) Glucose in the muscles
 - (C) Salt in the blood
 - (D) Oxygen in the blood
 - (E) Water in the blood
 - (F) Nitrogen in the lungs
 - (G) Carbon dioxide in the blood
- 93.** A rectangular hall is 12 m by 9 m. A carpet is laid leaving a uniform 1 m margin of bare floor all around. Find the area of the carpet. [\[Perimeter & Area\]](#)
- (A) 90 m²
 - (B) 63 m²
 - (C) 80 m²
 - (D) 108 m²
 - (E) 38 m²
 - (F) 88 m²
 - (G) 70 m²
- 94.** Three angles meet at a point and are 100 degrees, 80 degrees, and 95 degrees, with one more angle. Find the fourth angle. [\[Lines & Angles\]](#)
- (A) 85 deg
 - (B) 90 deg
 - (C) 105 deg
 - (D) 80 deg
 - (E) 95 deg
 - (F) 75 deg

95. The heavy rain that falls during a cyclone comes mainly from: [\[Winds, Storms & Cyclones\]](#)

- (A) Air pressure squeezing out water
- (B) Burning of gases in the clouds
- (C) Dust mixing with sunlight
- (D) Sea water thrown directly upward
- (E) Condensation of large amounts of rising water vapour
- (F) Cold air freezing the ground
- (G) Melting of high-altitude rocks

96. A frog can respire in two ways. While fully under water, a frog breathes mainly through its: [\[Respiration in Organisms\]](#)

- (A) Feathers
- (B) Dry scales
- (C) Tail fins
- (D) Lungs
- (E) Gills
- (F) Spiracles
- (G) Moist skin

97. A square garden has a side of 30 m with a circular pond of radius 7 m at its centre. Using $\pi = 22/7$, find the area of garden NOT covered by the pond. [\[Perimeter & Area\]](#)

- (A) 900 m²
- (B) 846 m²
- (C) 718 m²
- (D) 792 m²
- (E) 154 m²
- (F) 746 m²
- (G) 746 cm²

98. An angle is three times as large as its own complement. Find the angle. [\[Lines & Angles\]](#)

- (A) 60 deg
- (B) 67.5 deg
- (C) 75 deg
- (D) 22.5 deg
- (E) 30 deg
- (F) 135 deg

99. On hearing an official cyclone warning, the safest action is to: [\[Winds, Storms & Cyclones\]](#)

- (A) Stand near large windows
- (B) Drive toward the coast
- (C) Wait until water enters the house
- (D) Stay during the calm eye to inspect damage
- (E) Go outside to watch the sea
- (F) Move to a safer or higher place and ignore rumours
- (G) Spread unverified messages quickly

100. During a 3-minute sprint a runner breathes at 45 times per minute, compared with a resting 15 times per minute. How many EXTRA breaths does the sprint cause over those 3 minutes? [\[Respiration in Organisms\]](#)

- (A) 30
- (B) 180
- (C) 45
- (D) 135
- (E) 60
- (F) 90